KaxaNuk

Sharing knowledge

Strategy Presentation

Template

Content

- Strategy Definition
- Feature Engineering
- Strategy Design

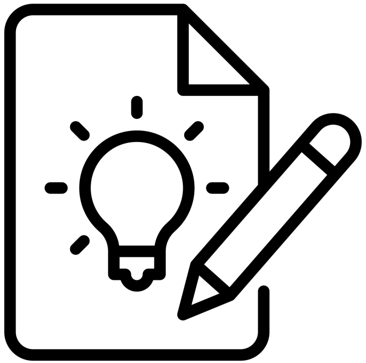
Strategy Definition

Subtitle

Idea Description

Explain the idea of the strategy and why you think it can be successful.

Try to be as specific as possible, the asset class you want to explore, and the specific money making idea.



Background Research

Check if similar ideas have been tested and validated.

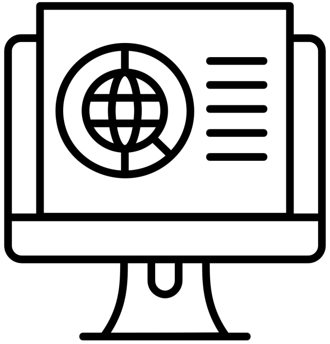
Paper Title

Link: https://www.kaxanuk.mx/

- Bullets that discuss the strategies ideas.

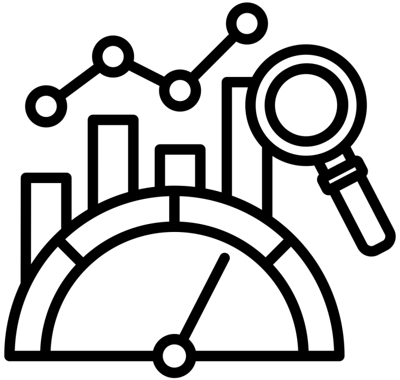
Investable Universe and Constraints

- US Country ETFs, Global Stocks,
- The tickers must have more than three months of trading history
- Market capitalization thresholds
- Liquidity requirements like minimum daily volume traded
- Asset class restrictions
- Geographic limitations
- Long only
- Long / short
- Cash only
- Leverage
- Margin
- Sector
- Factor exposure limits
- Benchmark tracking error



Benchmark

- Indices
- ACWI ETF
- Equal Weight Country ETFs
- SPDR S&P 500 ETF Trust
  - FMP Ticker - SPY
  - ISIN - US78462F1030
  - Ipo - 1/29/1993
  - Link - https://www.ssga.com/us/en/intermediary/etfs/spdr-sp-500-etf-trust-spy



Hypothesis

Market Inefficiency

- Price momentum exists due to investor underreaction to news
- Earnings announcement drift persists due to slow information diffusion

Risk Premium

- Companies with higher volatility should offer higher returns
- Illiquid stocks command a premium for holding risk

Fundamental

- Companies that are undervalued or overvalued

Behavioral

- Investors systematically overreact to negative news
- Retail investors chase past performance creating predictable patterns

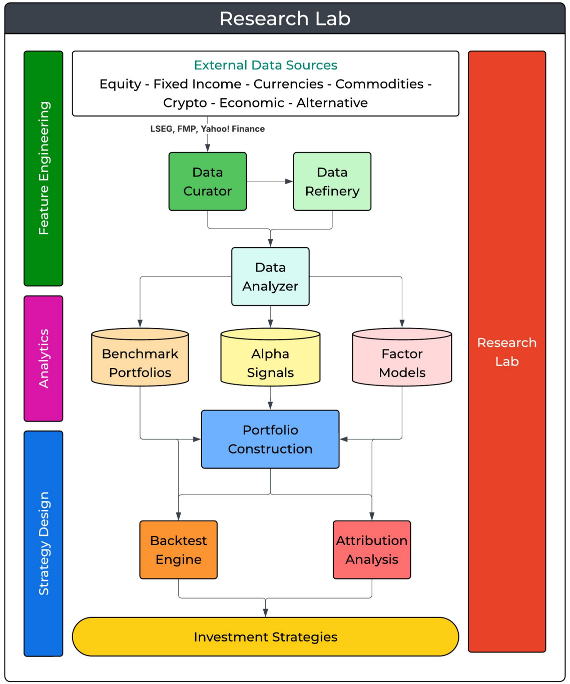
Structural

- Quarter-end rebalancing creates temporary price pressure
- Options expiration affects underlying stock volatility

Null Hypothesis

- Beat the benchmark

Unified Modeling Language Diagrams



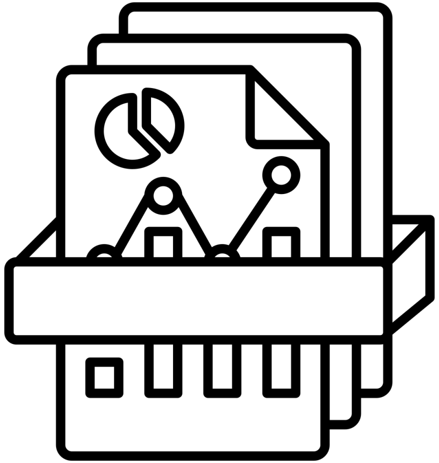
Feature Engineering

Subtitle

Data

The variables that we want to analyze:

- Market Data Open, High, Low, Close
- Fundamental Income, Balance Sheet, Cash Flows
- Analyst Estimates
- Earnings Transcripts
- Economic
- News



Data Analysis

Time Series Analysis

- Price trends and patterns
- Volatility clustering
- Seasonality and cyclical patterns
- Return distributions

Key Financial Metrics

- Returns (daily, monthly, annual)
- Risk measures (standard deviation, VaR, beta)
- Trading volumes
- Price-to-earnings ratios
- Market capitalization

Common Visualizations

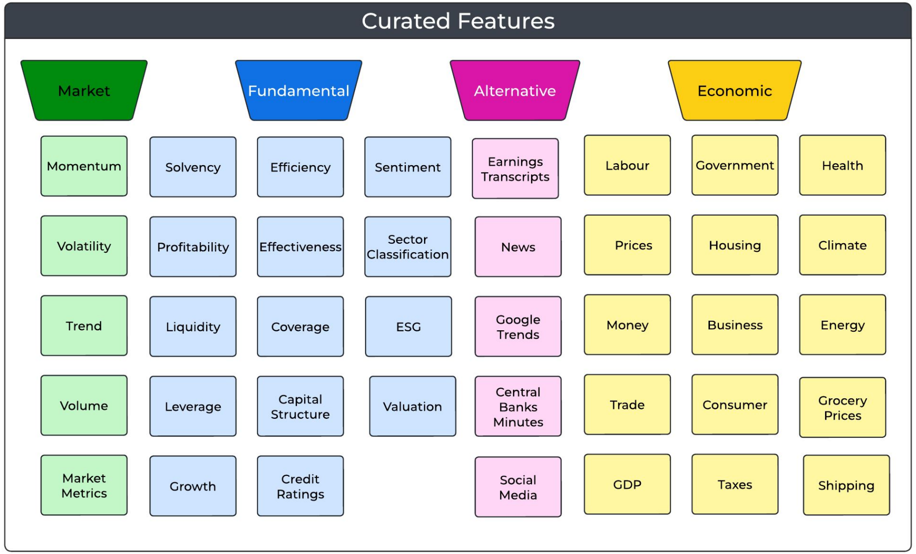
- Candlestick charts
- Moving averages
- Correlation heatmaps
- Box plots for return distributions
- Volume analysis

Features

Describe the features you will use:

- Momentum
- Volatility
- Value
- Quality
- Growth
- Sentiment
- Macro Signals

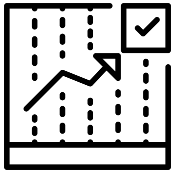
To avoid overfitting try to keep these features constants until the end of the research project.



Strategy Signals

Weighted Scoring

- Linear combinations
- Non-linear transformations
- Machine learning models
- Dynamic weighting schemes
- Regime-dependent weights

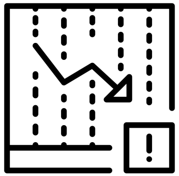


Probabilistic Frameworks

- Return distribution estimates
- Statistical confidence measures
- Bayesian updating methods
- Probability of outperformance
- Risk-adjusted forecasts

Ranking Approaches

- Percentile rankings
- Z-scores
- Quantile assignments
- Cross-sectional ordering
- Multi-factor ranks



Expected Return Models

- Factor return forecasts
- Risk premium adjustments
- Transaction cost estimates
- Decay rate modeling
- Capacity considerations

Strategy Design

Subtitle

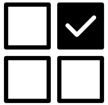
Strategy Modeling

Most research stops at:

"Does the signal predict returns?"

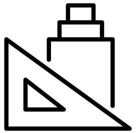
But strategy design requires answering three deeper questions during construction:

Selection



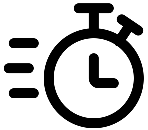
➔ Which securities enter the portfolio?

Sizing



➔ How much capital does each position receive?

Timing

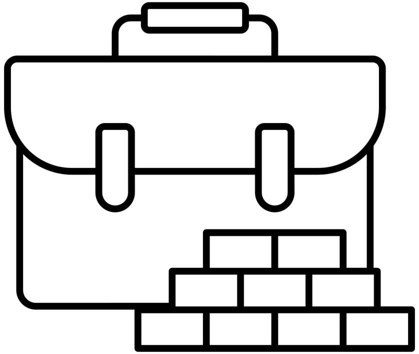


➔ When do we enter, rebalance, or exit?

Portfolio Construction

Taking into account the model logic we can construct historical portfolios under different approaches and constraints:

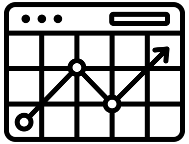
- ETF portfolios target 10-15 holdings
- Stocks portfolios target 30-40 holdings
- Revised weekly and dynamically rebalanced depending in threshold
- Rebalance holdings to correct asset allocation
- Rebalancing existing positions to return to “Target” weights
- Equal weight
- Mean-variance optimization
- Black-Litterman
- HRP
- Maximum or Minimum Weights Limits
- Sector Weights Limits



Backtesting

Performance Metrics

- Returns and risk-adjusted returns
- Maximum drawdown
- Volatility measures
- Trading frequency and turnover
- Position holding periods

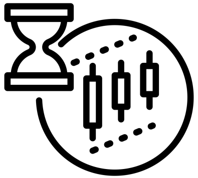


Strategy Robustness

- Out-of-sample performance
- Parameter sensitivity
- Market regime analysis
- Seasonality effects
- Diversification benefits

Implementation Feasibility

- Capacity analysis
- Transaction cost impact
- Operational requirements
- Capital needs
- Market accessibility

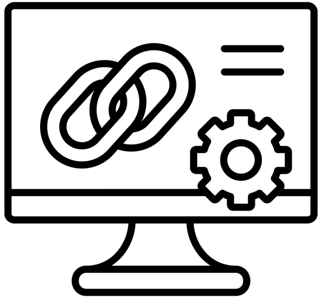


Strategy Evolution

- Adaptation to market changes
- Parameter optimization
- Overfitting assessment
- Improvement opportunities
- Lessons learned

Always Remember

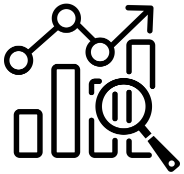
Keep track of your backtesting results to avoid overfitting and data mining, before making any change to the combination logic or portfolio construction and risk considerations complete the attribution analysis.



Attribution Analysis

Benchmark Comparison

- Absolute return differences
- Tracking error analysis
- Information ratio
- Active share metrics
- Risk-adjusted performance

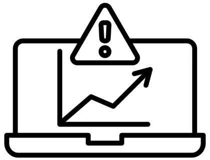


Security Selection

- Stock-specific returns
- Position sizing impact
- Entry/exit timing
- Selection skill within sectors
- High conviction positions

Sector and Industry Analysis

- Active sector weights vs benchmark
- Sector contribution to returns
- Sector timing decisions
- Industry rotation effects
- Sector correlation impacts



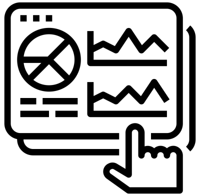
Factor Decomposition

- Style factor exposures
- Risk factor contributions
- Alpha factor effectiveness
- Factor timing
- Factor interaction effects

Attribution Analysis

Position-Level Controls

- Stop-loss triggers and execution
- Position size limits
- Sector / industry caps
- Concentration limits
- Entry / exit rules

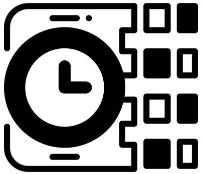


Drawdown Management

- Maximum drawdown limits
- Time under water rules
- Recovery targets
- Capital preservation rules
- De-risking triggers

Portfolio-Level Management

- Overall leverage limits
- Net exposure targets
- Beta neutrality requirements
- Factor exposure constraints
- Correlation controls



Stress Testing

- Historical scenario analysis
- Market regime testing
- Volatility shock scenarios
- Liquidity stress events
- Correlation breakdown tests

Scoring Template

Strategy Blueprint	Scoring Concept	Weight	Score (0–100)	Weighted Score	Gap (Score)	Description
Strategy Definition	Hypothesis & Source of Returns	11.11%	0	0.0	-11.1	Clarity and strength of the investment hypothesis, including a clear explanation of where returns are expected to come from.
	Research Foundation	11.11%	0	0.0	-11.1	Support for the idea through financial reasoning, prior research, or theoretical grounding.
	Scientific Process	11.11%	0	0.0	-11.1	How rigorously the idea is translated into a testable and validated framework.
Feature Engineering	Data Quality & Integrity	11.11%	0	0.0	-11.1	Reliability and correctness of the data used in the strategy.
	Feature Design & Relevance	11.11%	0	0.0	-11.1	How well features reflect the hypothesis and add meaningful information.
	Signal Construction	11.11%	0	0.0	-11.1	The process of transforming features into actionable signals.
Strategy Design	Portfolio Construction Logic	11.11%	0	0.0	-11.1	How signals are translated into capital allocation decisions.
	Backtesting & Robustness	11.11%	0	0.0	-11.1	Quality and credibility of the backtesting process.
	Attribution & Understanding	11.11%	0	0.0	-11.1	Ability to explain and decompose strategy performance.
Overall Score		**100.00%**		**0.0**		

Scoring Definitions	Score
Hypothesis & Source of Returns	
No clear hypothesis or source of returns	0
Vague idea with unclear return mechanism	33
Clear hypothesis with reasonable return explanation	66
Strong, well-defined hypothesis with a clear, testable source of returns	100
Research Foundation	
No supporting research or reasoning	0
Minimal references or weak justification	33
Some relevant research and logical support	66
Strong, well-integrated research and theoretical foundation	100
Scientific Process	
No structured testing or methodology	0
Partial structure, inconsistent testing	33
Clear testing process with some validation	66
Rigorous, reproducible, and well-documented research process	100

Scoring Definitions	Score
Data Quality & Integrity	
Data issues not considered	0
Basic awareness of data limitations	33
Data quality addressed with reasonable controls	66
Fully robust, clean, and bias-controlled dataset	100
Feature Design & Relevance	
No feature design or generic signals	0
Basic or standard features with limited relevance	33
Relevant and reasonably designed features	66
High-quality, differentiated, and well-justified features	100
Signal Construction	
No clear signal construction	0
Basic or unclear signal logic	33
Structured and reasonable signal generation	66
Robust, transparent, and stable signal framework	100

Scoring Definitions	Score
Portfolio Construction Logic	
No clear portfolio construction logic	0
Basic or incomplete allocation rules	33
Defined and reasonable portfolio construction	66
Well-structured, realistic, and disciplined allocation framework	100
Backtesting & Robustness	
No backtesting or unrealistic setup	0
Basic backtest with major limitations	33
Solid backtesting with some robustness checks	66
Rigorous, realistic, and robust validation process	100
Attribution & Understanding	
No attribution or explanation of results	0
Limited or superficial analysis	33
Reasonable breakdown of performance drivers	66
Comprehensive and clear understanding of return sources	100

Conclusions

Subtitle

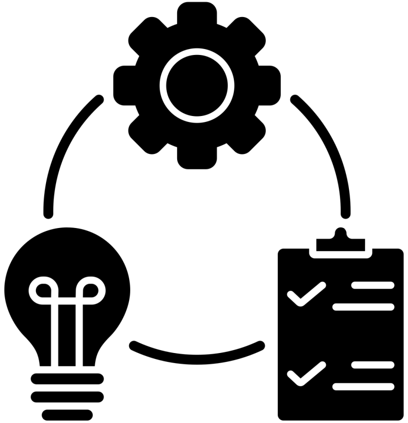
Hypothesis Validation

Understand the statistical implications and strategy consistency results.

Is This An Implementable Strategy?

Provide specific bullets to understand if the strategy should be implemented for paper trading.

- Factor decay is real - signals weaken over time
- Market impact can kill returns
- Capacity constraints matter
- Technology and infrastructure is critical
- Risk management is as important as alpha



Next Steps

Production Implementation

- Build robust production infrastructure
- Set up real-time data pipelines
- Implement automated monitoring systems
- Create failover procedures
- Develop live paper trading environment

Performance Monitoring

- Define key performance indicators (KPIs)
- Establish monitoring dashboards
- Create alert systems
- Set up regular review processes
- Track factor exposures

Strategy Enhancement

- Refine risk management framework
- Explore additional alpha factors
- Optimize execution algorithms
- Consider strategy combinations/ensembles
- Research regime adaptation methods

Business Considerations

- Calculate realistic capacity limits
- Develop scaling plans
- Establish capital raising strategy
- Consider operational requirements
- Plan for team expansion needs

Contact

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